

Texas A&M University STAT QE NOTE

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1 STAT 614 Advanced Probability

1.1 Chapter 1: Measure Theory

1.1.1 Probability Spaces

Key points:

- Probability space $(\Omega, \mathcal{F}, P)$
- σ -algebra $\mathcal{F}$
- measurable space: $(\Omega, \mathcal{F})$
- measure: a nonnegative countably additive set function
- If we are given a set Ω and a collection $\mathcal{A}$ of subsets of Ω , then there is a smallest σ -algebra containing $\mathcal{A}$. We will call this the σ -algebra generated by $\mathcal{A}$ and denote it by $\sigma(\mathcal{A})$.

1.1.2 Distributions

Key points:

- If X is a random variable, then X induces a probability measure on $\mathbf{R}$ called its **distribution** by setting $\mu(A) = P(X \in A)$ for Borel Sets A . Using the notation introduced previously, the right-hand side can be written as $P(X^{-1}(A)) = P(\{w : X(w) \in A\})$. In words, we pull $A \in \mathcal{R}$ back to $X^{-1}(A) \in \mathcal{F}$ and then take P of that set.

Exercise:

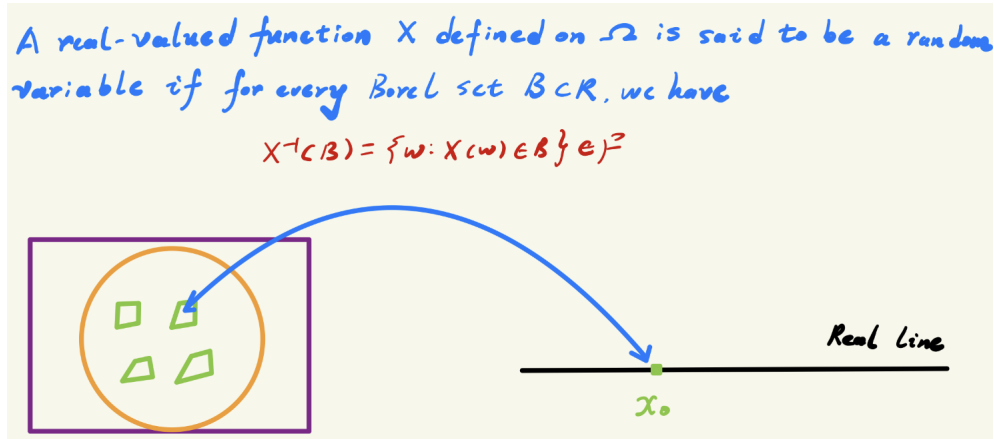
- **Theorem 1.2.6** For $x > 0$

$$(x^{-1} - x^{-3}) \exp(-x^2/2) \leq \int_x^\infty \exp(-y^2/2) dy \leq x^{-1} \exp(-x^2/2)$$

1.1.3 Random Variables

Key points:

- measurable map
A function $X : \Omega \rightarrow S$ is said to be a **measurable map** from $(\Omega, \mathcal{F})$ to $(S, \mathcal{S})$ if
$$X^{-1}(B) \equiv \{\omega : X(\omega) \in B\} \in \mathcal{F} \quad \text{for all } B \in \mathcal{S}$$
- random vector
- random variable: X is $\mathcal{F}$ -measurable



- **Theorem 1.3.1**

If $\{\omega : X(\omega) \in A\} \in \mathcal{F}$ for $A \in \mathcal{A}$ and $\mathcal{A}$ generates $\mathcal{S}$ (i.e., $\mathcal{S}$ is the smallest σ -field that contains $\mathcal{A}$), then X is measurable.

- Theorem 1.3.4

If $X : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$ and $f : (S, \mathcal{S}) \rightarrow (T, \mathcal{T})$ are measurable maps, then $f(X)$ is a measurable map from $(\Omega, \mathcal{F})$ to $(T, \mathcal{T})$

- Theorem 1.3.5

If $X_1, \dots, X_n$ are random variables and $f : (\mathbb{R}^n, \mathcal{R}^n) \rightarrow (\mathbb{R}, \mathcal{R})$ is measurable, then $f(X_1, \dots, X_n)$ is a random variable.

- Theorem 1.3.6

If $X_1, \dots, X_n$ are random variables, then $X_1 + \dots + X_n$ is a random variable.

- **Converges almost surely**

$$\Omega_o \equiv \{\omega : \lim_{n \rightarrow \infty} X_n \text{ exists}\} = \{\omega : \limsup_{n \rightarrow \infty} X_n - \liminf_{n \rightarrow \infty} X_n = 0\}$$

If $P(\Omega_o) = 1$, we say that x_n **converges almost surely**, or **a.s.** for short. This type of convergence is called **almost everywhere** in measure theory.

1.1.4 Integration

Key points:

- Integration Notation

1. When $(\Omega, \mathcal{F}, \mu) = (\mathbb{R}^d, \mathcal{R}^d, \lambda)$, we write $\int f(x) dx$ for $\int f d\lambda$.

2. When $(\Omega, \mathcal{F}, \mu) = (\mathbf{R}, \mathcal{R}, \lambda)$ and $E = [a, b]$, we write $\int_a^b f(x) dx$ for $\int_E f d\lambda$.
3. When $(\Omega, \mathcal{F}, \mu) = (\mathbf{R}, \mathcal{R}, \mu)$ with $\mu((a, b]) = G(b) - G(a)$ for $a < b$, we write $\int f(x) dG(x)$ for $\int f d\mu$.
4. When Ω is a countable set, $\mathcal{F} =$ all subsets of Ω , and μ is counting measure, we write $\sum_{i \in \Omega} f(i)$ for $\int f d\mu$.

1.1.5 Properties of the Integral

Key points:

- **Jensen's inequality**

- Norm of function: $\|f\|_p = (\int |f|^p d\mu)^{1/p}$ for $1 \leq p < \infty$

- **Hölder's inequality**

- **Cauchy-Schwarz inequality**

- **Convergence in measure/probability**

We say that $f_n \rightarrow f$ in **measure**, i.e., for any $\epsilon > 0$, $\mu(\{x : |f_n(x) - f(x)| > \epsilon\}) \rightarrow 0$ as $n \rightarrow \infty$. On a space of finite measure, this is a weaker assumption than $f_n \rightarrow f$ a.e.. Note here, we can regard $\mu(\{x : |f_n(x) - f(x)| > \epsilon\})$ as a sequence of measure. And the sign of **lim** is outside compared with the definition of **a.e.**. So on a space of finite measure, this is a weaker assumption than $f_n \rightarrow f$ **a.e.**.

- **Bounded convergence theorem**

Let E be a set with $\mu(E) < \infty$. Suppose f_n vanishes on E^c , $|f_n(x)| \leq M$, and $f_n \rightarrow f$ in measure. Then

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

- **Fatou's Lemma**

If $f_n \geq 0$, then

$$\liminf_{n \rightarrow \infty} \int f_n d\mu \geq \int \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu.$$

- **Monotone convergence theorem**

If $f_n \geq 0$ and $f_n \uparrow f$, then

$$\int f_n d\mu \uparrow \int f d\mu.$$

- **Dominated convergence theorem**

If $f_n \rightarrow f$ a.e., $|f_n| \leq g$ for all n , and g is integrable, then

$$\int f_n d\mu \rightarrow \int f d\mu.$$

Exercise:

- Poof of **Cauchy-Schwarz inequality**

1.1.6 Expected Value

Key points:

- Expectation: $EX = \int X dP$
- **Jensen's inequality**
- Norm of function: $\|X\|_r = (E|X|^r)^{1/r}$ for $1 \leq r < \infty$
- **Hölder's inequality**
- **Chebyshev's inequality**
- **Markov's inequality**
- **Fatou's Lemma**
- **Monotone convergence theorem**
- **Dominated convergence theorem:**
The special case of this theorem is **bounded convergence theorem**

1.1.7 Product Measures, Fubini's Theorem

Key points:

- **Fubini's Theorem** If $f \geq 0$ or $\int |f| d\mu < \infty$, then

$$(*) \quad \int_X \int_Y f(x, y) \mu_2(dy) \mu_1(dx) = \int_{X \times Y} f d\mu = \int_Y \int_X f(x, y) \mu_1(dx) \mu_2(dy)$$

1.2 Chapter 2: Law of Large Numbers

1.2.1 Independence

Theorem 2.1.8 *In order for $X_1, \dots, X_n$ to be independent, it is sufficient that for all $x_1, \dots, x_n \in (-\infty, \infty]$*

$$P(X_1 \leq x_1, \dots, X_n \leq x_n) = \prod_{i=1}^n P(X_i \leq x_i)$$

Theorem 2.1.16 *Suppose that X with density f and Y with distribution function G are independent. Then $Z = X + Y$ has density*

$$h(z) = \int f(z - y) dG(y)$$

When Y has density g , the last formula can be written as

$$h(z) = \int f(z - y) g(y) dy$$

1.2.2 Weak Laws of Large Numbers

Lemma 2.2.2 *If $p > 0$ and $\mathbb{E}|Z_n|^p \rightarrow 0$, then $Z_n \rightarrow 0$ in probability.*

Proof Chebyshev's inequality, Theorem 1.6.4, with $\varphi(x) = x^p$ and $X = |Z_n|$ implies that if $\epsilon > 0$, then

$$P(|Z_n| \geq \epsilon) \leq \epsilon^{-p} \mathbb{E}|Z_n|^p \rightarrow 0. \quad \square$$

Theorem 2.2.3 (L^2 weak law) *Let $X_1, X_2, \dots$ be uncorrelated random variables with $\mathbb{E}X_i = \mu$ and $\text{var}(X_i) \leq C < \infty$. If $S_n = X_1 + \dots + X_n$, then as $n \rightarrow \infty$, $S_n/n \rightarrow \mu$ in L^2 and in probability.*

Theorem 2.2.14 *Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}|X_i| < \infty$. Let $S_n = X_1 + \dots + X_n$ and let $\mu = \mathbb{E}X_1$. Then $S_n/n \rightarrow \mu$ in probability.*

1.2.3 Borel-Cantelli lemma

Theorem 2.3.2 *$X_n \rightarrow X$ in probability if and only if for every subsequence $X_{n(m)}$, there is a further subsequence $X_{n(m_k)}$ that converges almost surely to X .*

Theorem 2.3.4 *If f is continuous and $X_n \rightarrow X$ in probability, then $f(X_n) \rightarrow f(X)$ in probability. If, in addition, f is bounded, then $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$.*

1.2.4 Strong Law of Large Numbers

Theorem 2.4.1 (Strong law of large numbers) *Let $X_1, X_2, \dots$ be pairwise independent identically distributed random variables with $\mathbb{E}|X_i| < \infty$. Let $\mathbb{E}X_i = \mu$ and $S_n = X_1 + \dots + X_n$. Then $S_n/n \rightarrow \mu$ a.s. as $n \rightarrow \infty$.*

1.3 Chapter 3: Central Limit Theorems

1.3.1 The De Moivre-Laplace Theorem

- **The De Moivre-Laplace Theorem**
- **Stirling's formula** $n! \sim n^n e^{-n} \sqrt{2\pi n}$ as $n \rightarrow \infty$

1.3.2 Weak Convergence

- **Definition of weak convergence**

A sequence of distribution functions is said to **converge weakly** to a limit F (written $F_n \Rightarrow F$) if $F_n(y) \rightarrow F(y)$ for all y that are continuity points of F . A sequence of random variables X_n is said to **converge weakly** or **converge in distribution** to a limit X_∞ (written $X_n \Rightarrow X_\infty$) if their distribution functions $F_n(x) = P(X_n \leq x)$ converge weakly.

- **Skorohod's Representation Theorem (Thm 3.2.8)**

If $F_n \Rightarrow F_\infty$, then there are random variables Y_n , $1 \leq n \leq \infty$, with distribution F_n so that $Y_n \rightarrow Y_\infty$ a.s.

Theorem 3.2.9

$X_n \Rightarrow X_\infty$ if and only if for every bounded continuous function g we have $Eg(X_n) \rightarrow Eg(X_\infty)$.

Theorem 3.2.10 (Continuous Mapping Theorem) Let g be a measurable function and $D_g = \{x : g \text{ is discontinuous at } x\}$. If $X_n \xrightarrow{d} X_\infty$ and $P(X_\infty \in D_g) = 0$, then $g(X_n) \xrightarrow{d} g(X_\infty)$. If in addition g is bounded, then $\mathbb{E}[g(X_n)] \rightarrow \mathbb{E}[g(X_\infty)]$.

1.3.3 Characteristic Functions

- **Definition of C.F.**
 $\phi(t) = \mathbb{E}[e^{itX}] = \mathbb{E}[\cos(tX)] + i\mathbb{E}[\sin(tX)]$

- **Theorem 3.3.1**

All Characteristic functions have the following properties:

- $\phi(0) = 1$,
- $\phi(-t) = \overline{\phi(t)}$,
- $|\phi(t)| = |Ee^{itX}| \leq E|e^{itX}| = 1$
- $|\phi(t+h) - \phi(t)| \leq E|e^{ihX} - 1|$, so $\phi(t)$ is uniformly continuous on $(-\infty, \infty)$.
- $Ee^{it(aX+b)} = e^{itb}\phi(at)$

- **Theorem 3.3.14**

If $\int |\phi(t)| dt < \infty$, then μ has bounded continuous density $f(y) = \frac{1}{2\pi} \int e^{-ity} \phi(t) dt$

- **Theorem 3.3.17 (Continuity theorem)**

Let μ_n , $1 \leq n \leq \infty$ be probability measures with ch.f. φ_n . (i) If $\mu_n \Rightarrow \mu_\infty$, then $\varphi_n(t) \rightarrow \varphi_\infty(t)$ for all t . (ii) If $\varphi_n(t)$ converges pointwise to a limit $\varphi(t)$ that is continuous at 0, then the associated sequence of distributions μ_n is tight and converges weakly to the measure μ with characteristic function φ .

- **Theorem 3.3.18**

If $\int |x|^n \mu(dx) < \infty$, then its characteristic function φ has a continuous derivative of order n given by $\varphi^{(n)}(t) = \int (ix)^n e^{itx} \mu(dx)$.

1.3.4 Central Limit Theorems

- **Theorem 3.4.1** Let $X_1, X_2, \dots$ be i.i.d. with $EX_i = \mu$, $\text{var}(X_i) = \sigma^2 \in (0, \infty)$. If $S_n = X_1 + \dots + X_n$, then

$$(S_n - n\mu)/\sigma n^{1/2} \Rightarrow \chi$$

where χ has the standard normal distribution.

1.4 Chapter 4: Conditional Expectation

1.4.1 Concepts

- Given are a probability space $(\Omega, \mathcal{F}_o, P)$, a σ -field $\mathcal{F} \subset \mathcal{F}_o$, and a random variable $X \in \mathcal{F}_o$ with $E|X| < \infty$. We define the **conditional expectation** of X given $\mathcal{F}$, $E(X|\mathcal{F})$, to be any random variable Y that has

- (i) $Y \in \mathcal{F}$, i.e., is $\mathcal{F}$ measurable
- (ii) for all $A \in \mathcal{F}$,

$$\int_A X dP = \int_A Y dP$$

Any Y satisfying (i) and (ii) is said to be a **version** of $E(X|\mathcal{F})$. The first thing to be settled is that the conditional expectation exists and is unique. We tackle the second claim first, but start with a technical point.

- **Lemma 4.1.1** *If Y satisfies (i) and (ii), then it is integrable*
- **Uniqueness** If Y' also satisfies (i) and (ii), then

$$\int_A Y dP = \int_A Y' dP \quad \text{for all } A \in \mathcal{F}$$

Taking $A = \{Y - Y' \geq \epsilon > 0\}$, we see

$$0 = \int_A X - X dP = \int_A Y - Y' dP \geq \epsilon P(A)$$

so $P(A) = 0$. Since this holds for all ϵ , we have $Y \leq Y'$ a.s., and interchanging the roles of Y and Y' , we have $Y = Y'$ a.s. Technically, all equalities such as $Y = E(X|\mathcal{F})$ should be written as $Y = E(X|\mathcal{F})$ a.s., but we have ignored this point in previous chapters and will continue to do so.

- **Theorem 4.1.2**
If $X_1 = X_2$ on $B \in \mathcal{F}$, then $E(X_1|\mathcal{F}) = E(X_2|\mathcal{F})$ a.s. on B .

1.4.2 Properties

- **Theorem 4.1.9**
In the first two parts we assume $E|X|, E|Y| < \infty$.
(a) *Conditional expectation is linear:*

$$E(aX + Y|\mathcal{F}) = aE(X|\mathcal{F}) + E(Y|\mathcal{F})$$

- (b) *If $X \leq Y$, then*

$$E(X|\mathcal{F}) \leq E(Y|\mathcal{F})$$

- (c) *If $X_n \geq 0$ and $X_n \uparrow X$ with $EX < \infty$, then*

$$E(X_n|\mathcal{F}) \uparrow E(X|\mathcal{F})$$

- **Theorem 4.1.10**

If φ is convex and $E|X|, E|\varphi(X)| < \infty$, then

$$\varphi(E(X|\mathcal{F})) \leq E(\varphi(X)|\mathcal{F})$$

- **Theorem 4.1.11**

Conditional expectation is a contraction in L^p , $p \geq 1$.

- **Theorem 4.1.12**

If $\mathcal{F} \subset \mathcal{G}$ and $E(X|\mathcal{G}) \in \mathcal{F}$, then $E(X|\mathcal{F}) = E(X|\mathcal{G})$.

- **Theorem 4.1.13** If $\mathcal{F}_1 \subset \mathcal{F}_2$, then

$$(i) \quad E(E(X|\mathcal{F}_1)|\mathcal{F}_2) = E(X|\mathcal{F}_1)$$

$$(ii) \quad E(E(X|\mathcal{F}_2)|\mathcal{F}_1) = E(X|\mathcal{F}_1).$$

- **Theorem 4.1.14**

If $X \in \mathcal{F}$ and $E|Y|, E|XY| < \infty$, then

$$E(XY|\mathcal{F}) = XE(Y|\mathcal{F}).$$

- **Theorem 4.1.15**

Suppose $EX^2 < \infty$. $E(X|\mathcal{F})$ is the variable $Y \in \mathcal{F}$ that minimizes the “mean square error”

$$E(X - Y)^2.$$

2 STAT 613 Statistical Methodology I: Frequency Inference

2.1 Module 1: Review

- Moment Generating Function
- Transformation of Random Variables
- Location and scale families
- Properties of Normal distribution
- Order Statistics: Marginal distribution of $X_{(i)}$ and Joint distribution of $X_{(i)}$ and $X_{(j)}$
- Convergence concepts
- Slutsky's Theorem
- Weak Law of Large Numbers and Central Limit Theorem
- First and Second Order Delta Method

2.2 Module 2: Point Estimation

- Sufficient Statistics
- Factorization Theorem
- Minimal Sufficient Statistics: Definition + Theorem
- Ancillary Statistics + Location & Scale family
- Complete Statistics: If a statistic is complete and sufficient, then it has to be minimally sufficient.
- Basu's Theorem
- UMVUE: uniqueness/uncorrelated with all unbiased estimators of 0.
- Rao-Blackwell Theorem
- Law of total Expectation + Law of total Variance
- Lehman-Scheffe Theorem to find UMVUE
- Rao-Blackwell + Lehman-Scheffe to find UMVUE
- Fisher Information
- Cramer-Rao Inequality + CRLB to find UMVUE

- Exponential family:

$$f(x | \boldsymbol{\theta}) = h(x) c(\boldsymbol{\theta}) \exp \left(\sum_{j=1}^k w(\theta_j) t_j(x) \right)$$

where $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_k)$. Then the statistic

$$T(\mathbf{X}) = \left(\sum_{i=1}^n t_1(X_i), \sum_{i=1}^n t_2(X_i), \dots, \sum_{i=1}^n t_k(X_i) \right)$$

is complete as long as the parameter space Θ contains an open set in $\mathbb{R}^k$.

- Curved exponential family:

A *curved exponential family* is a family of densities of the form above for which the dimension of the vector $\boldsymbol{\theta}$ is equal to $d < k$. If $d = k$, the family is a *full exponential family*.

2.3 Module 3: Hypothesis Testing

- Rejection region & Power function

- Neyman-Pearson Lemma

Consider testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$, where the pdf or pmf corresponding to θ_i is $f(x | \theta_i)$ for $i = 0, 1$. We use a test with rejection region $\mathcal{R}$ that satisfies

$$\phi(x) = \begin{cases} 1 & \text{if } f(x | \theta_1) > k f(x | \theta_0), \\ 0 & \text{if } f(x | \theta_1) < k f(x | \theta_0), \end{cases}$$

for some $k \geq 0$, and the test's level is $\alpha = P_{\theta_0}(X \in \mathcal{R})$.

(a) Any test that satisfies the above is the most powerful level α test.

(b) If there exists a test satisfying the above, then every MP level- α test is a size- α test and every MP level- α test satisfies the above except perhaps on a set A satisfying $P_{\theta_0}(X \in A) = P_{\theta_1}(X \in A) = 0$.

- Uniformly Most Powerful test

- Monotone likelihood ratio

- p-value: $p(X) = \inf\{\alpha : X \in \mathcal{R}_\alpha\}$.

- UMPU Test

A test ϕ is said to be an *unbiased* level- α test for $H_0 : \theta \in \Theta_0$ VS. $H_1 : \theta \in \Theta_1$ if:

- (i) $\beta(\theta) \leq \alpha$ for all $\theta \in \Theta_0$,

(ii) $\beta(\theta) \geq \alpha$ for all $\theta \in \Theta_1$,

where $\beta(\theta) = \mathbb{E}_\theta[\phi(X)]$ is the power function of the test (the probability of rejecting H_0 when the true parameter is θ).

Note: The above is the complete definition for Unbiased Test; the below is the definition for UMPU Test.

If the densities are s.t. all tests have continuous power functions, then if there exists a UMP test T among the tests satisfying $E(T') = \alpha$ whenever θ belongs to the boundary of Θ_0 and Θ_1 , then T is UMPU level α test.

- **UMPU Test More**

To test

$$H_0 : \eta = \eta_0 \quad \text{vs.} \quad H_1 : \eta \neq \eta_0,$$

the test function $\phi(T(X))$ is given by

$$\phi(T(X)) = \begin{cases} 1 & \text{if } T(X) < c_1 \text{ or } T(X) > c_2, \\ 0 & \text{otherwise.} \end{cases}$$

Here, the constants c_1 and c_2 are chosen so that making $\phi(T(X))$ the UMPU (Uniformly Most Powerful Unbiased) level- α test.

With a bit of calculation, one can see that if $T(X)$ is symmetrically distributed about a point a under H_0 , then from $E_{\eta_0}[\phi(T(X))] = \alpha$ and $c_1 + c_2 = 2a$, one can solve for c_1 and c_2 .

- **LRT**

Note: No matter how many parameters you have in the question, you can still use **MLE** to maximize the Likelihood (take log first). So, the testing question was transferred to a ratio between Restricted and Unrestricted Likelihood optimization questions.

2.4 Module 4: Interval Estimation

- **Confidence coefficient**

EX1: $X \sim \mathcal{N}(\mu, \sigma^2)$, σ^2 is known

$$\inf_{\theta} P_{\theta} \left(\theta \in \left[\bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \right] \right) = 1 - \alpha$$

EX2: $X \sim \text{Binomial}(n = 10, p)$

X takes value in $0, \dots, 10$. Consider the CI: $\left[\max \left(0, \frac{X-1}{10} \right), \min \left(1, \frac{X+1}{10} \right) \right]$
Finally,

$$P_p \left(p \in \left[\max \left(0, \frac{X-1}{10} \right), \min \left(1, \frac{X+1}{10} \right) \right] \right) = \sum_{x \in S(p)} \binom{10}{x} p^x (1-p)^{10-x}$$

where $S(p)$ is the set of all x satisfying

$$\frac{x-1}{10} \leq p \leq \frac{x+1}{10}.$$

EX3:

When working with confidence *intervals*, we occasionally will work with more general sets. When working in general, and not being quite sure of the exact form of our sets, we will speak of confidence *sets*. A confidence set with confidence coefficient equal to some value, say $1 - \alpha$, is simply called a $1 - \alpha$ *confidence set*.

• **Construct-CI I: Inverting a Test Statistic**

Example (Inverting a normal test)

Let $X_1, \dots, X_n$ be iid $n(\mu, \sigma^2)$ and consider testing $H_0: \mu = \mu_0$ versus $H_1: \mu \neq \mu_0$. For a fixed α level, a UMPU test has rejection region $\{\mathbf{x}: |\bar{x} - \mu_0| > z_{\alpha/2}\sigma/\sqrt{n}\}$. Note that H_0 is accepted for sample points with $|\bar{x} - \mu_0| \leq z_{\alpha/2}\sigma/\sqrt{n}$ or, equivalently,

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu_0 \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

Since the test has size α , this means that $P(H_0 \text{ is rejected} \mid \mu = \mu_0) = \alpha$ or, stated in another way, $P(H_0 \text{ is accepted} \mid \mu = \mu_0) = 1 - \alpha$. Combining this with the above characterization of the acceptance region, we can write

$$P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu_0 \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \mid \mu = \mu_0\right) = 1 - \alpha.$$

But this probability statement is true *for every* μ_0 . Hence, the statement

$$P_\mu\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

is true. The interval $[\bar{x} - z_{\alpha/2}\sigma/\sqrt{n}, \bar{x} + z_{\alpha/2}\sigma/\sqrt{n}]$, obtained by *inverting* the acceptance region of the level α test, is a $1 - \alpha$ confidence interval.

These sets are connected to each other by:

$$(x_1, \dots, x_n) \in A(\mu_0) \Leftrightarrow \mu_0 \in C(x_1, \dots, x_n).$$

• **Construct-CI I: General Algorithm**

For each $\theta_0 \in \Theta$, let $A(\theta_0)$ be the acceptance region of a level α test of $H_0: \theta = \theta_0$. For each $\mathbf{x} \in \mathcal{X}$, define a set $C(\mathbf{x})$ in the parameter space by

$$C(\mathbf{x}) = \{\theta_0 : \mathbf{x} \in A(\theta_0)\}. \quad (1)$$

Then the random set $C(\mathbf{X})$ is a $1 - \alpha$ confidence set. Conversely, let $C(\mathbf{X})$ be a $1 - \alpha$ confidence set. For any $\theta_0 \in \Theta$, define

$$A(\theta_0) = \{\mathbf{x} : \theta_0 \in C(\mathbf{x})\}.$$

Then $A(\theta_0)$ is the acceptance region of a level α test of $H_0: \theta = \theta_0$.

Reject region $\longrightarrow$ Acceptance region $\longrightarrow$ Invert test statistic $\longrightarrow$ Confidence interval

- **Construct-CI II: Pivotal Quantities**

A function $Q(\underline{x}, \theta)$ of data and parameters is called a pivot if the distribution of $Q(\underline{x}, \theta)$ is free of any unknown parameter.

- **Shortest-CI I:**

$$\begin{aligned} \textcircled{1} \quad & \int_a^b f(x) dx = 1 - \alpha \\ \textcircled{2} \quad & f(a) = f(b) > 0 \\ \textcircled{3} \quad & a \leq x^* \leq b \end{aligned}$$

where x^* is the mode of the distribution of $f(x)$, then $[a, b]$ is the shortest interval satisfying $\textcircled{1}$.

- **Shortest-CI II:** Our objective is to minimize $b - a$ subject to

$$\int_a^b f(x) dx = 1 - \alpha.$$

This is a constrained optimization, and also note that the optimized value of b is a function of a , denote it by $b(a)$. Clearly we have to minimize two equations:

$$\frac{db}{da} - 1 = 0, \quad \frac{db}{da} f(b) - f(a) = 0.$$

They together give $f(b) = f(a)$.

- **False Coverage**

Note that the probability of true coverage for an interval $C(\mathbf{X})$ is given by $P_\theta(\theta \in C(\mathbf{X}))$, i.e., the probability of covering the true parameter. The probability of *false* coverage is the function of θ, θ' such that

$$P_\theta(\theta' \in C(\mathbf{X})), \theta \neq \theta', \quad \text{if } C(\mathbf{X}) = [L(\mathbf{X}), U(\mathbf{X})]$$

$$P_\theta(\theta' \in C(\mathbf{X})), \theta < \theta', \quad \text{if } C(\mathbf{X}) = (-\infty, U(\mathbf{X})]$$

$$P_\theta(\theta' \in C(\mathbf{X})), \theta > \theta', \quad \text{if } C(\mathbf{X}) = [L(\mathbf{X}), \infty)$$

it is the probability of covering θ' when the true parameter is θ . Note that false coverage will be big if we unnecessarily cover unimportant θ' , and it would potentially increase the length of the interval without increasing coverage. Therefore, our aim should be to reduce false coverage.

- **UMA confidence set**

A $1 - \alpha$ confidence set that minimizes the probability of false coverage over a class of $1 - \alpha$ confidence sets is called a *uniformly most accurate* (UMA) confidence set. Thus, for example, we would consider looking for a UMA confidence set among sets of the form $[L(x), \infty)$. UMA confidence sets are constructed by inverting the acceptance regions of UMP tests, as we will prove below.

- **Theorem: Connection between UMP and false coverage**

Let $\mathbf{X} \sim f(\mathbf{x}|\theta)$, where θ is a real-valued parameter. For each $\theta_0 \in \Theta$, let $A^*(\theta_0)$ be the UMP level α acceptance region of a test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$. Let $C^*(\mathbf{x})$ be the $1 - \alpha$ confidence set formed by inverting the UMP acceptance regions. Then for any other $1 - \alpha$ confidence set C ,

$$P_\theta(\theta' \in C^*(\mathbf{X})) \leq P_\theta(\theta' \in C(\mathbf{X})) \quad \text{for all } \theta' < \theta.$$

Proof: Consider $(x_1, \dots, x_n) \in A(\theta) \Leftrightarrow \theta \in C(x_1, \dots, x_n)$.

Example

We have previously seen that for the normal problem $C(\bar{X}) = \{\mu : \mu \geq \bar{X} - z_\alpha \sigma / \sqrt{n}\}$ is the acceptance region for the UMP test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu > \mu_0$. Therefore, this interval reduces the probability of false coverage.

- **Unbiased confidence set**

A $1 - \alpha$ confidence set $C(\mathbf{x})$ is *unbiased* if

$$P_\theta(\theta' \in C(\mathbf{X})) \leq 1 - \alpha \quad \text{for all } \theta \neq \theta'.$$

Explanation: Thus, for an unbiased confidence set, the probability of false coverage is never more than the **minimum probability of true coverage** (link to the confidence coefficient).

- **Theorem: Connection between Unbiased (including UMPU) and false coverage**

The acceptance region of an unbiased test can be inverted to construct an unbiased confidence set.

Proof: Consider the case: $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$ (Assignment6 Q5).

- **Theorem: Connection between length of CI and probability of false coverage**

Let X be a real-valued random variable with $X \sim f(x|\theta)$, where θ is a real-valued parameter. Let $C(x) = [L(x), U(x)]$ be a confidence interval for θ . If $L(x)$ and $U(x)$ are both increasing functions of x , then for any value θ^* ,

$$\mathbb{E}_{\theta^*}(\text{Length}[C(\mathbf{X})]) = \int_{\theta \neq \theta^*} P_{\theta^*}(\theta \in C(\mathbf{X})) d\theta.$$

Proof:

$$\begin{aligned}
E_{\theta^*}(\text{Length}[C(X)]) &= \int_{\mathcal{X}} \text{Length}[C(X)] f_{\theta^*}(x) dx \\
&= \int_{\mathcal{X}} [U(x) - L(x)] f_{\theta^*}(x) dx \\
&= \int_{\mathcal{X}} \left[\int_{L(x)}^{U(x)} d\theta \right] f_{\theta^*}(x) dx \\
&= \int_{\Theta} \left[\int_{U^{-1}(\theta)}^{L^{-1}(\theta)} f_{\theta^*}(x) dx \right] d\theta \\
&= \int_{\Theta} P_{\theta^*}(U^{-1}(\theta) \leq X \leq L^{-1}(\theta)) d\theta \\
&= \int_{\Theta} P_{\theta^*}(\theta \in C(X)) d\theta = \int_{\theta \neq \theta^*} P_{\theta^*}(\theta \in C(X)) d\theta
\end{aligned}$$

Explanation: This theorem shows that there is a formal relationship between the length of a confidence interval and its probability of false coverage. In the two-sided case, this implies that minimizing the probability of false coverage carries along some guarantee of length optimality. In the one-sided case, however, the analogy does not quite work.

2.5 Module 5: True & False

- Any one-to-one function of a sufficient statistic is also sufficient.
- Any one-to-one function of a minimal sufficient statistic is also minimal sufficient.
- Any one-to-one function of a complete statistics is also complete.
- Minimal sufficient statistics for a parameter are unique.
- A minimal sufficient statistic is complete.
- The minimal sufficient statistic for a one-dimensional parameter has to be one-dimensional.
- A minimal sufficient statistic is independent of an ancillary statistic.
- $N(\mu, \sigma^2)$, $\mu \in R$, $\sigma^2 > 0$: $(\bar{x}, \sum_{i=1}^n X_i^2)$ or $(\bar{x}, s^2)$ is sufficient & complete.
- $N(\mu, \mu^2)$, $\mu \in R$: $(\bar{x}, \sum_{i=1}^n X_i^2)$ or $(\bar{x}, s^2)$ is sufficient but not complete.
- $[a, b]$ is confidence interval, but $[a, b] \cup [c, d]$ is not.
- In LRT, the denominator is optimized in the whole sample space.

- The more common two-sided interval,

$$C(\bar{x}) = \left\{ \mu : \bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right\},$$

is not UMA, since it is obtained by inverting the two-sided acceptance region from the test of $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$, hypotheses for which no UMP test exists.

- The Acceptance region of the UMP test has the minimum false coverage.
- Acceptance region of an unbiased test leads to an unbiased confidence interval.
- Expected length of a CI is related to the prob of false coverage.

3 STAT 632 Statistical Methodology II: Bayesian Statistics

3.1 Introduction

Exchangeability

If a set of random variables $X_1, X_2, \dots, X_n$ satisfies:

$$P(X_1, X_2, \dots, X_n) = P(X_{\pi(1)}, X_{\pi(2)}, \dots, X_{\pi(n)})$$

for any permutation π , then these random variables are **exchangeable**.
In simple terms:

- If you shuffle the order of the data, the joint distribution of the data does not change, then the data is exchangeable.
- Exchangeability does not require each data point to be independently and identically distributed (i.i.d....), but it is a weaker generalization of i.i.d.
- Note, exchangeability means if you shuffle the label of data, for example, $x_i|\theta_i$ changed to $x_j|\theta_i$, the joint distribution remains the same.

Order Statistics

If we consider the joint probability density function (PDF) of all n order statistics $X_{(1)}, X_{(2)}, \dots, X_{(n)}$, its expression is given by:

$$f_{X_{(1)}, X_{(2)}, \dots, X_{(n)}}(x_1, x_2, \dots, x_n) = n! \prod_{i=1}^n f(x_i)$$

which holds for $x_1 < x_2 < \dots < x_n$; otherwise, the density is zero.

This result indicates that, after sorting the sample, the joint density is simply an adjustment of the original i.i.d sample joint density, ensuring that the order statistics have an appropriate normalization constant (Probability summation is equal to 1).

If we want to assign a probability to the event of interest, we should consider the whole sample space we described. Here, the joint probability density function (PDF) of all n order statistics equals the $n!$ times summation of $\prod_{i=1}^n f(x_i)$. Since we have $n!$ times permutations produce the same order statistics.

3.2 Exact Inference

3.2.1 Prior Distributions

- Noninformative Prior (including Improper Prior)
- Informative Prior
- Conjugate Prior

Family	Conjugate Prior
Binomial(N, θ)	$\theta \sim \text{beta}(\alpha, \lambda)$
Poisson(θ)	$\theta \sim \text{gamma}(\delta_0, \gamma_0)$
$N(\mu, \sigma^2)$, σ^2 known	$\mu \sim N(\mu_0, \sigma_0^2)$
$N(\mu, \sigma^2)$, μ known, $\tau = 1/\sigma^2$	$\tau \sim \text{gamma}(\delta_0, \gamma_0)$
$\text{gamma}(\alpha, \lambda)$, α known	$\lambda \sim \text{gamma}(\delta_0, \gamma_0)$
$\text{Beta}(\alpha, \lambda)$, λ known	$\alpha \sim \text{gamma}(\delta_0, \gamma_0)$

- Jeffreys Prior

Definition: Fisher Information

Let $p(x | \theta)$ denote the density of x given θ . The Fisher information is

$$I(\theta) = -E \left[\frac{\partial^2 \log p(x | \theta)}{\partial \theta^2} \right].$$

If θ is a $p \times 1$ vector, then

$$I(\theta) = -E \left[\frac{\partial^2 \log p(x | \theta)}{\partial \theta_i \partial \theta_j} \right]_{p \times p}.$$

and thus $I(\theta)$ is a $p \times p$ matrix.

Definition

Jeffreys's prior is defined as

$$\pi(\theta) \propto |I(\theta)|^{\frac{1}{2}},$$

where $|\cdot|$ denotes determinant.

3.2.2 Predictive Distributions

$$\begin{aligned} p(z | x) &= \int_{\Theta} p(z | \theta) p(\theta | x) d\theta \\ &= E_{\theta|x} [p(z | \theta)]. \end{aligned}$$

3.3 Bayesian Hypothesis Testing

3.3.1 Posterior Ratio

Before data (we need to assign probability to H_0 and H_1):

$$P(H_0) = P(H_a) = \frac{1}{2}$$

After collecting data, the posterior probabilities:

$$P(H_0 | x) = \frac{P(x | H_0) \cdot P(H_0)}{P(x)}$$

$$P(H_a | x) = \frac{P(x | H_a) \cdot P(H_a)}{P(x)}$$

Posterior ratio:

$$\frac{P(H_0 | x)}{P(H_a | x)} = \underbrace{\frac{P(x | H_0)}{P(x | H_a)}}_{\text{likelihood ratio}} \cdot \underbrace{\frac{P(H_0)}{P(H_a)}}_{\text{ratio of prior probs.}}$$

If

$$\frac{P(H_0 | x)}{P(H_a | x)} \geq 1$$

we prefer H_0 , reject H_a .

If

$$\frac{P(H_0 | x)}{P(H_a | x)} < 1$$

we prefer H_a , reject H_0 .

3.3.2 Bayes Factor

$$BF = \frac{p(H_0 | D)/p(H_0)}{p(H_1 | D)/p(H_1)}$$

$$= \frac{\left(\frac{p(H_0 | D)}{p(H_1 | D)} \right)}{\left(\frac{p(H_0)}{p(H_1)} \right)}$$

$$= \frac{p(D | H_0)}{p(D | H_1)}.$$

3.3.3 Bayes Factor Hypothesis Test

More formally, suppose our parameter space is Θ , and we wish to test

$$H_0 : \theta \in \Theta_{H_0}$$

$$H_1 : \theta \in \Theta_{H_1},$$

where

$$\Theta = \Theta_{H_0} \cup \Theta_{H_1} \quad \text{and} \quad \Theta_{H_0} \cap \Theta_{H_1} = \emptyset.$$

Then, the Bayes factor in favor of H_0 is defined as

$$BF = \frac{p(D | H_0)}{p(D | H_1)} = \frac{\int_{\Theta_{H_0}} p(D | \theta, H_0) \pi(\theta | H_0) d\theta}{\int_{\Theta_{H_1}} p(D | \theta, H_1) \pi(\theta | H_1) d\theta}$$

where Θ_{H_0} is the parameter space under H_0 , and Θ_{H_1} is the parameter space under H_1 .

3.3.4 Highest Posterior Density Regions

Definition Let $p(\theta | x)$ denote the posterior density of θ . A region R in the parameter space Θ is called an HPD region of content $1 - \alpha$ if

a)

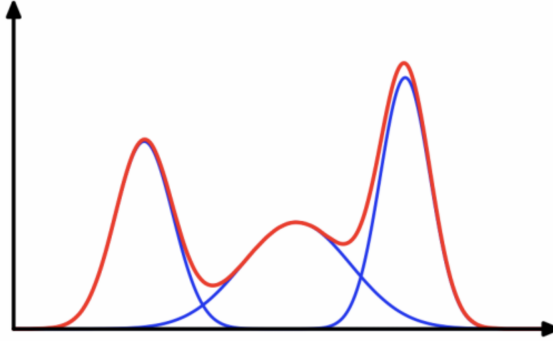
$$P(\theta \in R | x) = 1 - \alpha$$

b) for $\theta_1 \in R$ and $\theta_2 \notin R$,

$$p(\theta_1 | x) \geq p(\theta_2 | x).$$

Some Properties of HPD regions

1. It follows from the definition that for a given probability content $1 - \alpha$, the HPD region has the smallest possible volume in the parameter space Θ .
2. If we make the assumption that $p(\theta | x)$ is non-uniform over every region in the space of θ , then the HPD region of content $1 - \alpha$ is unique. Further, if θ_1 and θ_2 are two points such that $p(\theta_1 | x) = p(\theta_2 | x)$, then these two points are simultaneously included (or excluded) by a $1 - \alpha$ HPD region. The converse is also true.



That is, if $p(\theta_1 | x) \neq p(\theta_2 | x)$, then there exists a $1 - \alpha$ HPD region which includes one point but not the other.

Remark: In real-world distributions used for describing data, they are usually multimodal. So, from the algorithm point of view, you can think about a horizontal line parallel to the x-y plane and put it down until you get the region with probability $1 - \alpha$. That region is the HPD region you want.

3.4 Approximate Inference

3.4.1 Bayesian Central Limit Theorem

Suppose $x_1, \dots, x_n$ are independent observations from $p(x | \theta)$. Further, suppose that the posterior distribution is proper and its mode exists. Then as $n \rightarrow \infty$,

$$\theta | x \rightarrow N_p[\hat{\theta}_m, H^{-1}(\hat{\theta}_m)],$$

where $\hat{\theta}_m$ is the posterior mode of θ , obtained by solving

$$\frac{\partial}{\partial \theta_j} \log p^*(\theta | x) = 0,$$

Where $p^*(\theta | x) = p(x | \theta)\pi(\theta)$ is the kernel of the posterior density (the unnormalized posterior).

$$H(\theta) = - \left[\frac{\partial^2 \log p^*(\theta | x)}{\partial \theta_i \partial \theta_j} \right]$$

is a $p \times p$ matrix. The quantity $-H(\theta)$ is called the **Hessian matrix**. Thus, the asymptotic covariance matrix of θ is minus the inverse of the Hessian matrix, evaluated at $\hat{\theta}_m$;

$$H^{-1}(\hat{\theta}_m) = H^{-1}(\theta) \Big|_{\theta=\hat{\theta}_m}.$$

This is an asymptotic approximation, valid when $n \rightarrow \infty$.

3.4.2 Laplace Approximation

- **Laplace's method**

Suppose f is a smooth positive function of θ , and h is a smooth function of θ with $-h$ having unique maximum $\hat{\theta}$. For simplicity, take θ to be 1-dimensional. Suppose we wish to approximate

$$I = \int f(\theta) e^{-nh(\theta)} d\theta.$$

Using Taylor's series expansions of both f and h about $\hat{\theta}$

$$\hat{I} \approx e^{-nh(\hat{\theta})} f(\hat{\theta}) \sqrt{\frac{2\pi}{nh''(\hat{\theta})}}$$

If θ is p -dimensional, Laplace's method yields

$$\hat{I} = f(\hat{\theta}) \left(\frac{2\pi}{n} \right)^{p/2} |\tilde{\Sigma}|^{\frac{1}{2}} \exp[-nh(\hat{\theta})],$$

where

$$\tilde{\Sigma} = \left[\frac{\partial^2 h(\theta)}{\partial \theta_j \partial \theta_k} \right]^{-1} \Big|_{\theta=\hat{\theta}}.$$

- **First order approximation**

To this end, suppose we wish to compute the posterior expectation of a function $g(\theta)$, where we think of $-nh(\theta)$ as the unnormalized posterior density, that is,

$$-nh(\theta) = \log[p(x | \theta)\pi(\theta)],$$

Then

$$E[g(\theta)] = \frac{\int g(\theta) e^{-nh(\theta)} d\theta}{\int e^{-nh(\theta)} d\theta},$$

So

$$E[g(\theta)] = g(\hat{\theta}), \quad h(\hat{\theta}) = 0.$$

- **Second order approximation**

$$\begin{aligned} E[g(\theta)] &= \frac{\int g(\theta) e^{-nh(\theta)} d\theta}{\int e^{-nh(\theta)} d\theta} \\ &= \frac{\int e^{\log(g(\theta)) - nh(\theta)} d\theta}{\int e^{-nh(\theta)} d\theta} \\ &= \frac{\int e^{-nh^*(\theta)} d\theta}{\int e^{-nh(\theta)} d\theta} \end{aligned}$$

Now we apply Laplace's method with $f = 1$ in both the numerator and denominator.

$$E[g(\theta)] = \frac{|\Sigma^*|^{\frac{1}{2}} \exp[-nh^*(\theta^*)]}{|\tilde{\Sigma}|^{\frac{1}{2}} \exp[-nh(\hat{\theta})]},$$

where θ^* is the maximum of $-h^*$ and

$$\Sigma^* = \left[\frac{\partial^2 h^*(\theta)}{\partial \theta_j \partial \theta_k} \right]^{-1} \Big|_{\theta=\theta^*}.$$

3.4.3 Non-iterative Monte Carlo Methods

Direct Sampling

Goal: $\gamma = \mathbb{E}[f(\theta)] = \int f(\theta) p(\theta)$

Then if $\theta_1, \dots, \theta_N$ are i.i.d. samples from $p(\theta)$, we have

$$\hat{\gamma} = \frac{1}{N} \sum_{j=1}^N f(\theta_j),$$

which converges to $\mathbb{E}[f(\theta)]$ with probability 1 as $N \rightarrow \infty$, by the strong law of large numbers.

But the problem is that sometimes it's hard for us to directly sample from the posterior distribution.

Indirect Sampling

Goal:

$$\begin{aligned}\text{Approximate } E[f(\theta)|x] &= \frac{\int f(\theta)L(\theta)\pi(\theta)d\theta}{\int L(\theta)\pi(\theta)d\theta} \\ &= \int f(\theta) \frac{L(\theta)\pi(\theta)}{\int L(\theta)\pi(\theta)d\theta} d\theta\end{aligned}$$

Since, in the high-dimensional setting, it's hard to do the integration for the denominator part, which means it's hard to get the accurate posterior density function. That's the motivation why we need to develop some indirect methods.

- **Importance sampling**

Suppose we can roughly approximate the normalized likelihood times the prior (accurate posterior density), $\frac{L(\theta)\pi(\theta)}{c}$, by some density $g(\theta)$, from which we can easily sample.

Then, defining the weight function

$$w(\theta) = \frac{L(\theta)\pi(\theta)}{g(\theta)}$$

We have

$$\begin{aligned}E[f(\theta)|x] &= \frac{\int f(\theta) w(\theta) g(\theta) d\theta}{\int w(\theta) g(\theta) d\theta} \\ &\approx \frac{\frac{1}{N} \sum_{j=1}^N f(\theta_j) w(\theta_j)}{\frac{1}{N} \sum_{j=1}^N w(\theta_j)}\end{aligned}$$

where the θ_j are i.i.d samples from $g(\theta)$. Here, $g(\theta)$ is called the importance function.

Note: How closely $g(\theta)$ resembles $\frac{L(\theta)\pi(\theta)}{c}$ (i.e., the normalized posterior) controls how good the approximation is. When $g(\theta) = \frac{L(\theta)\pi(\theta)}{c}$, it means $g(\theta)$ is exactly the posterior we want.

- **Rejection sampling**

- **The weighted bootstrap**

Based on the importance sampling, we have $\theta_1, \dots, \theta_N$ from some approximating density $g(\theta)$. Then, define $w_i = \frac{L(\theta_i)\pi(\theta_i)}{g(\theta_i)}$, $q_i = \frac{w_i}{\sum_{i=1}^N w_i}$.

Now draw θ^* from the **discrete** distribution over $\theta_1, \dots, \theta_N$, which places mass q_i at θ_i .

Then θ^* is a sample from

$$p(\theta|x) = \frac{L(\theta)\pi(\theta)}{\int L(\theta)\pi(\theta)d\theta}$$

with the approximation improving as $N \rightarrow \infty$.

Proof

$$\begin{aligned}
P(\theta^* \leq a) &= \sum_{i=1}^N q_i I(-\infty < \theta_i \leq a) \\
&= \frac{\frac{1}{N} \sum_{i=1}^N w_i I(-\infty < \theta_i \leq a)}{\frac{1}{N} \sum_{i=1}^N w_i} \\
&\xrightarrow{N \rightarrow \infty} \frac{\mathbb{E}_g \left[\frac{L(\theta) \pi(\theta)}{g(\theta)} I(-\infty < \theta \leq a) \right]}{\mathbb{E}_g \left[\frac{L(\theta) \pi(\theta)}{g(\theta)} \right]} \\
&= \frac{\int_{-\infty}^a \left[\frac{L(\theta) \pi(\theta)}{g(\theta)} \right] g(\theta) d\theta}{\int_{-\infty}^{\infty} \left[\frac{L(\theta) \pi(\theta)}{g(\theta)} \right] g(\theta) d\theta} \\
&= \frac{\int_{-\infty}^a L(\theta) \pi(\theta) d\theta}{\int_{-\infty}^{\infty} L(\theta) \pi(\theta) d\theta} \\
&= \int_{-\infty}^a p(\theta | x) d\theta,
\end{aligned}$$

so that θ^* is now approximately distributed as $p(\theta | x)$, as desired.

Similar to importance and rejection sampling, we need $g(\theta) \approx p(\theta | x)$, or else a very large N will be required to obtain acceptable accuracy.

3.4.4 Iterative Monte Carlo Methods

- Gibbs Sampling
- Metropolis Algorithm
- Metropolis-Hastings Algorithm

3.4.5 MCMC

A major concern of MCMC theory is to determine the conditions under which there exists an invariant (stationary) distribution π and conditions under which iterations of the transition kernel converge to the invariant distribution. The invariant distribution π satisfies

$$\pi(Y) = \int_{\mathbb{R}^d} P(Y|X) \pi(X) dX$$

The discrete version is

$$\sum_{X \in S} \pi(X) P(Y|X) = \pi(Y) \quad \text{for all } Y \in S$$

MCMC: In the MCMC method, the invariant/stationary/target density is known (up to a constant multiple). The known target density form is like $\pi(\tilde{\Theta}|D) \propto f(D|\tilde{\Theta}) \cdot \pi(\tilde{\Theta})$. It is the target density π from which samples are desired. However, the transition kernel is unknown. To generate samples from $\pi(\cdot)$, the methods find and utilize a transition kernel $P(Y|X)$ whose n th iterate converges to $\pi(\cdot)$ for large n . The problem is to find an appropriate $P(Y|X)$. We may put some restriction on the transition kernel $P(\cdot)$.

Methods: We have two methods to prove the picked/designed transition kernel $P(Y|X)$ is what we want! One is using the Reversibility condition (one appropriate condition for transition kernel). The other is directly using the definition.

- Reversibility or detailed balanced distribution

A Markov chain with TK $P(\cdot)$ satisfies the detailed balance condition with stationary density π satisfying

$$P(X|Y)\pi(Y) = P(Y|X)\pi(X)$$

for every (X, Y) . It can be interpreted as saying that the rate at which the system moves from X to Y is the same at which the system moves from Y to X . Now, if $P(Y|X)$ satisfies the reversibility condition, then $\pi(\cdot)$ is an invariant density of $P(X, \cdot)$.

We can prove that if the transition kernel satisfies this reversibility condition, then that's what we want.

Since $P(B|X) = P(Y \in B|X) = \int_B P(Y|X)dY$

$$\begin{aligned} \int_X P(B|X)\pi(X)dX &= \int_X \int_B P(Y|X)\pi(X)dXdY \\ &= \int_X \int_B P(X|Y)\pi(Y)dXdY \\ &= \int_B \pi(Y)dY \end{aligned}$$

- Using the definition for invariant/target distribution for π .

Examples of reversibility distribution

EX1: Metropolis-Hastings Algorithm

Given $X^{(t)}$

- Generate Y_t from the proposal distribution $q(Y_t|X^{(t)})$
- Define $\alpha(X^{(t)}, Y_t) = \min \left\{ \frac{\pi(Y)q(X|Y)}{\pi(X)q(Y|X)}, 1 \right\}$

- Take $X^{(t+1)} = Y_t$ with probability $\alpha(X^{(t)}, Y_t)$
- $X^{(t+1)} = X^{(t)}$ with probability $1 - \alpha(X^{(t)}, Y_t)$

The TK for the M-H algorithm is

$$P(Y|X) = \alpha(X, Y)q(Y|X) + [1 - \alpha(X, Y)]\delta_X(Y)$$

where $r(X) = \int \alpha(X, Y)q(Y|X)dY$ and $\delta_X(Y) = 1$ if $X = Y$ and 0 otherwise.

[EX2: Reversibility of M-H sampler](#)

Suppose $\alpha(X, Y) < 1$, hence $\alpha(Y, X) = 1$

$$\begin{aligned}\alpha(X, Y)q(Y|X)\pi(X) &= \frac{\pi(Y)q(X|Y)}{\pi(X)q(Y|X)}q(Y|X)\pi(X) \\ &= \alpha(Y, X)q(X|Y)\pi(Y)\end{aligned}$$

Similarly it can be shown $(1 - r(x))\delta_X(Y)\pi(X) = (1 - r(Y))\delta_Y(X)\pi(Y)$, which together established the detailed balance for M-H algorithm.

Examples of using the definition for invariant distribution for π

[EX1: Gibbs Sampling Algorithm](#)

Transition kernel for Gibbs sampler. Let the two states be

$$X = [X_1, \dots, X_k]$$

$$Y = [Y_1, \dots, Y_k]$$

Now,

$$\begin{aligned}P(Y|X) &= \pi(Y_1|X_2, \dots, X_k)\pi(Y_2|Y_1, X_3, \dots, X_k) \\ &= \dots, \pi(Y_k|Y_1, \dots, Y_{k-1})\end{aligned}$$

This is not reversible. Hence, we have to show that it provides you with the invariant distribution. We prove it for 2 dimensions, but the extension is simple.

For $k = 2$

$$X = [X_1, X_2]$$

$$Y = [Y_1, Y_2]$$

Now,

$$\begin{aligned}\pi(Y) &= \int P(Y|X)\pi(X)dx \\ &= \int_{X_1} \int_{X_2} \pi(Y_1|X_2)\pi(Y_2|Y_1)\pi(X_1, X_2)dx_1dx_2 \\ &= \pi(Y_2|Y_1) \int_{X_1} \int_{X_2} \pi(Y_1|X_2)\pi(X_1, X_2)dx_1dx_2 \\ &= \pi(Y_2|Y_1) \int_{X_2} \pi(Y_1|X_2)\pi(X_2)dx_2 \\ &= \pi(Y_2|Y_1)\pi(Y_1) \\ &= \pi(Y_1, Y_2)\end{aligned}$$

[EX2: Grouped and Collapsed Gibbs sampler](#)

Grouped Gibbs sampler In this algorithm, we first group some elements and draw them jointly from the **conditional** posterior distribution.

$$(i) (X_1, X_2) \sim \pi(X_1, X_2 | X_3)$$

$$(ii) X_3 \sim \pi(X_3 | X_1, X_2)$$

Transition kernel for modified collapsed Gibbs sampler. Let two states be

$$X = [X_1, X_2, X_3]$$

$$Y = [Y_1, Y_2, Y_3]$$

Now, $P(Y|X) = P(Y_1, Y_2 | X_3)P(Y_3 | Y_1, Y_2)$

$$\begin{aligned} \pi(Y) &= \int_{X_1} \int_{X_2} \int_{X_3} P(Y|X) \pi(X) dX \\ &= \int_{X_1} \int_{X_2} \int_{X_3} P(Y_1, Y_2 | X_3) P(Y_3 | Y_1, Y_2) \pi(X) dX \\ &= P(Y_3 | Y_1, Y_2) \int_{X_1} \int_{X_2} \int_{X_3} P(Y_1, Y_2 | X_3) \pi(X_1, X_2, X_3) dX \\ &= P(Y_3 | Y_1, Y_2) \int_{X_3} P(Y_1, Y_2 | X_3) \pi(X_3) dX_3 \\ &= \pi(Y_1, Y_2, Y_3) \end{aligned}$$

Collapsed Gibbs sampler In this algorithm, we first group some elements and draw them jointly from the **Marginal** posterior distribution.

$$(i) (X_1, X_2) \sim \pi(X_1, X_2)$$

$$(ii) X_3 \sim \pi(X_3 | X_1, X_2)$$

Now, $P(Y|X) = \pi(Y_1, Y_2) \pi(Y_3 | Y_1, Y_2)$

$$\begin{aligned} \pi(Y) &= \int_{X_1} \int_{X_2} \int_{X_3} P(Y|X) \pi(X) dX \\ &= \pi(Y_1, Y_2) \pi(Y_3 | Y_1, Y_2) \int_{X_1} \int_{X_2} \int_{X_3} \pi(X_1, X_2, X_3) dX \\ &= \pi(Y_1, Y_2) \pi(Y_3 | Y_1, Y_2) \\ &= \pi(Y_1, Y_2, Y_3) \end{aligned}$$

Modified collapsed Gibbs sampler In this algorithm, we first group some elements and draw them jointly from the **Marginal** posterior distribution. To draw this marginal joint distribution, we used mini-Gibbs steps.

$$(i) \quad X_1 \sim \pi(X_1|X_2)$$

$$(i) \quad X_2 \sim \pi(X_2|X_1)$$

$$(ii) \quad X_3 \sim \pi(X_3|X_1, X_2)$$

Now, $P(Y|X) = \pi(Y_1|X_2)\pi(Y_2|Y_1)\pi(Y_3|Y_1, Y_2)$

$$\begin{aligned} \pi(Y) &= \int P(Y|X)\pi(X)dx \\ &= \int_{X_1} \int_{X_2} \int_{X_3} \pi(Y_1|X_2)\pi(Y_2|Y_1)\pi(Y_3|Y_1, Y_2)\pi(X_1, X_2, X_3)dX_1dX_2dX_3 \\ &= \pi(Y_2|Y_1)\pi(Y_3|Y_1, Y_2) \int_{X_2} \pi(Y_1|X_2)\pi(X_2)dX_2 \\ &= \pi(Y_2|Y_1)\pi(Y_3|Y_1, Y_2)\pi(Y_1) \\ &= \pi(Y_1, Y_2, Y_3) \end{aligned}$$

3.4.6 Variational Inference

ELBO Optimization

$$\begin{aligned} \min_{q \in \mathcal{Q}} KL(q(\theta)||p(\theta|x)) &= \min_{q \in \mathcal{Q}} \mathbb{E}_{\theta \sim q} \left[\log \frac{q(\theta)}{p(\theta|x)} \right] \\ &= \min_{q \in \mathcal{Q}} \int_{-\infty}^{\infty} \log \left(\frac{q(\theta)}{p(\theta|x)} \right) q(\theta) d\theta. \end{aligned}$$

$$\begin{aligned} KL(q(\theta)||p(\theta|x)) &= \int_{\mathbb{R}} \log \left[\frac{q(\theta)}{p(\theta|x)} \right] q(\theta) d\theta \\ &= \int_{\mathbb{R}} \log \left[\frac{q(\theta)p(x)}{p(\theta, x)} \right] q(\theta) d\theta \\ &= \int_{\mathbb{R}} \log \left[\frac{q(\theta)}{p(\theta, x)} \right] q(\theta) d\theta + \int_{\mathbb{R}} \log p(x) q(\theta) d\theta \\ &= - \int_{\mathbb{R}} \log \left[\frac{p(\theta, x)}{q(\theta)} \right] q(\theta) d\theta + \log p(x). \end{aligned}$$

$\log p(x)$: Evidence

$$KL \geq 0 \iff \int_{-\infty}^{\infty} \log \left[\frac{p(\theta, x)}{q(\theta)} \right] q(\theta) d\theta \leq \log p(x)$$

$$\text{ELBO (Evidence lower Bound)} = \int_{-\infty}^{\infty} \log \left[\frac{p(\theta, x)}{q(\theta)} \right] q(\theta) d\theta$$

$$KL[q(\theta)||p(\theta|x)] = -\text{ELBO} + \log p(x)$$

$$\min_{q \in \mathcal{Q}} KL[q(\theta)||p(\theta|x)] \quad \text{equiv.} \quad \max_{q \in \mathcal{Q}} \text{ELBO}$$

Mean-field VB

$$\begin{aligned}\theta &\rightarrow \theta_1, \theta_2, \theta_3. \\ q(\theta) &= q_1(\theta_1)q_2(\theta_2)q_3(\theta_3).\end{aligned}$$

Here, the order of the fraction is different from the definition. Since we take one minus sign out of the expectation.

$$\begin{aligned}\text{ELBO} &= -\mathbb{E}_{\theta \sim q} \log \left[\frac{\prod_{i=1}^3 q_i(\theta_i)}{p(\theta, x)} \right] \\ &= \int [\log p(\theta, x) - \log q_1(\theta_1) - \log q_2(\theta_2) - \log q_3(\theta_3)] \prod_{i=1}^3 q_i(\theta_i) d\theta_1 d\theta_2 d\theta_3.\end{aligned}$$

If we want to maximize the ELBO, we can take the partial derivative of the ELBO concerning different $q_i^*(\theta_i)$.

Equally,

$$\begin{aligned}q_i^*(\theta_i) &\propto \exp \{ \mathbb{E}_{-\theta_i} \log p(\theta; x) \}, \\ &\propto \exp \{ \mathbb{E}_{-\theta_i} ((\log p(\theta_i | \theta_{-i}; x) + p(\theta_{-i}; x)) \}, \\ &\propto \exp \{ \mathbb{E}_{-\theta_i} (\log p(\theta_i | \theta_{-i}; x)) \} \quad \forall i = 1, 2, 3\end{aligned}$$

3.5 Tricky Part

The question workflow is:

- Look at the data
- Fit the model
- Find the likelihood

Attention:

- No coding was tested in the qualifying exam.
- Note, we should master the normal, beta, and gamma integrals.

References

- [1] Lawrence R. **Principe**, *The Secrets of Alchemy* (Chicago 2013).
- [2] Sometimes, people have extra things to say.